

12	A particle of mass 5.0×10^{-3} kg performing simple harmonic motion of amplitude 150 mm takes 47 s to make 50 oscillations. What is the maximum kinetic energy of the particle?
	A 2.0×10^{-3} J B 2.5×10^{-3} J C 3.9×10^{-3} J D 5.0×10^{-3} J
	Answer: B Frequency = 50/47 Hz $KE_{\max} = \frac{1}{2}m(\omega x_0)^2 = \frac{1}{2}m(2\pi f x_0)^2$ $= (0.5)(5.0 \times 10^{-3})(2\pi (50/47)(150 \times 10^{-3}))^2$ $= 2.5 \times 10^{-3} \text{ J}$